RISHABH KUMAR

Senior Data Scientist | AI & Behavioral Modeling | Deep Learning & LLM Calibration

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Senior Data Scientist and Machine Learning Engineer with **7+ years of experience** building production AI/ML systems across identity, behavioral modeling, audience intelligence, and representation learning. Specialized in **foundation representation learning, transformer architectures (cross-attention, mask-aware), synthetic consumer simulation, and empirical calibration** across **350M+ users**. Published author (**ICML 2025 Best Paper**, NeurIPS 2025 Workshop) on modeling user behavior from adaptive surveys. Proven record modernizing legacy systems and shipping scalable deep learning pipelines on AWS SageMaker and Apache Spark.

TECHNICAL EXPERTISE & COMPETENCIES

AI & Deep Learning: Transformers (Cross-Attention, Mask-Aware, Self-Attention), Representation Learning, Self-Supervised Learning, Autoencoders, Deep Matrix Factorization, Multi-Label Classification, Probabilistic Modeling, PU Learning, Anomaly Detection.

Generative AI & Agents: Synthetic Audience Simulation, Agent Persona Parameterization, LLM Calibration (Isotonic Regression, Conformal Prediction, Distribution Alignment), LLMOps & Evaluation, In-Context Persona Grounding, LangChain/LangGraph, RAG.

Languages & Tools: Python, PyTorch, TensorFlow/Keras, Scala, SQL/PostgreSQL, R, Hugging Face, Spark ML, XGBoost, LightGBM, scikit-learn, Optuna, MLflow, TensorBoard, Docker, Git/GitHub, CI/CD.

Big Data & Cloud Systems: Apache Spark, AWS Ecosystem (SageMaker, EMR, Batch, Step Functions, S3, EC2, Lambda, SQS, RDS), Grafana, PII Masking, Consent-Based Consumer Data Pipelines, Privacy-Preserving Identity Resolution.

PROFESSIONAL EXPERIENCE

Resonate Networks
Herndon, VA (Hybrid / Boston, MA)

Senior Data Scientist `Apr 2024 – Present`

- **Technical Lead for Keystone:** Served as technical lead for Keystone, coordinating a cross-functional team of 6 across data engineering, product, DS/ML, and executive stakeholders to drive technical direction and execution, moving identity intelligence from transient ID-level behavior toward durable person-level representations.
- **Self-Supervised Behavioral Autoencoder (BAE):** Developed a self-supervised transformer/autoencoder foundation model deployed on AWS SageMaker, compressing geo, domain, and NLP-derived category signals into **512-dimensional behavioral embeddings** across millions of daily events for **350M+ users**.
- **Large-Scale Survey Imputation & Persona Reconstruction:** Built a downstream supervised Deep Matrix Factorization model that consumes BAE embeddings to predict **~19K survey attributes (+0.07 mean AUC lift)**, reconstructing sparse adaptive survey responses with population-level calibration across 570K respondents — powering core audience segmentation products and campaign targeting.
- **ICML 2025 Best Paper Architecture (LANTERN):** Formulated and productionized a modular late-fusion cross-attention neural architecture fusing sparse user survey priors with external behavioral features via learned bimodal gating; achieved **+5 point recall lift** and strong performance on rare consumer preferences, directly preserving population heterogeneity.
- **Media Bubble & Ideological Cohort Clustering:** Built *Media Bubble*, an unsupervised behavioral clustering pipeline classifying **~20M devices** into ideological consumption cohorts using real-time behavioral event signals, directly informing campaign creative messaging and voter landscape strategy.

ML Engineer → ML Engineer II → Senior ML Engineer `Nov 2019 – Mar 2024`

- **Lookalike Audience Discovery (Positive-Unlabeled Learning):** Originated a new lookalike-audience product by adapting an existing pipeline for a new use case, using positive-unlabeled learning to identify high-propensity prospects across **~500M HEM profiles**; outperformed production in **~90% of evaluated audiences** on recall-lift and cumulative-gain metrics.
- **Distributed Spark/Scala NLP Pipeline Modernization:** Migrated a legacy MapReduce NLP pipeline to a distributed Apache Spark/Scala architecture processing millions of daily internet events; reduced compute costs by **~35%** and accelerated processing runtime by **~2.8x**.
- **Large-Scale Production Orchestration:** Streamlined and productionized automated training, inference, and monitoring workflows across AWS SageMaker, EMR, AWS Batch, S3, Step Functions, and SQS for **300M+ active user profiles** across ~1,800 XGBoost models.

HiLabs Inc.
Germantown, MD

Data Scientist `Mar 2019 – Oct 2019`

- Developed risk adjustment and anomaly detection products for healthcare data using frequent-pattern mining on **~3.5B records**, translating high-dimensional data into strategic insights and client-facing recommendations.

PUBLICATIONS

Modeling User Behavior from Adaptive Surveys with Supplemental Context **Best Paper Award**
Aman Shukla, Daniel Patrick Scantlebury, Rishabh Kumar — **ICML 2025 New in ML Workshop** (arXiv:2507.20919)

Responsible Imputation of User Behavior Surveys via Mask-Aware Transformers `NeurIPS 2025`
NeurIPS 2025 Workshop on Reliable ML from Unreliable Data

EDUCATION

Carnegie Mellon University (CMU) Pittsburgh, PA | 2017 – 2018
BITS Pilani — B.E. Chemical Engineering & M.Sc. Biological Sciences (Dual Degree) Pilani, India | 2011 – 2016

HONORS & AWARDS

Best Paper Award: ICML 2025 New in ML Workshop (July 2025)
Reed Award Winner: Best Use of Data Analytics / ML in Online Fundraising (Feb 2026)
1st Prize Winner: Resonate Networks Annual Hackathon (June 2021 & July 2025)